

PROJECT

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SECURE SECRETS WITH SECRETS MANAGER

SECURITY SERVICE

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Objective

This project uses AWS Secrets Manager to securely manage application credentials instead of hardcoding them. It highlights how cloud-native secret management strengthens security and simplifies credential handling.

A simple web app lists S3 buckets using stored credentials, showing how applications can stay secure, scalable, and compliant with best practices.

Services

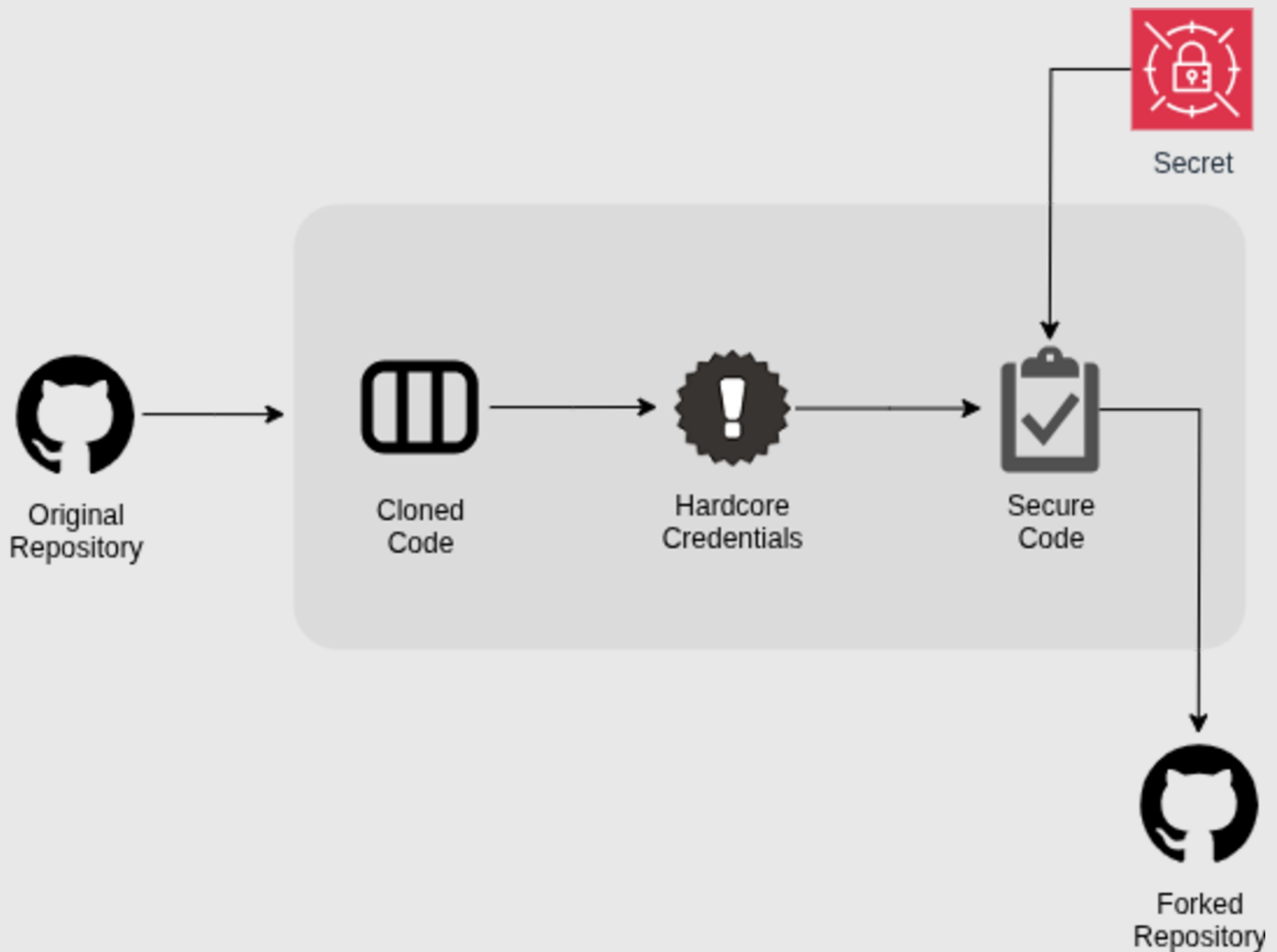
AWS Secrets Manager - Securely stores and retrieves secrets like API keys or credentials with rotation and access control.

Amazon S3 - Scalable object storage used by the web app to list buckets.

AWS IAM - Provides secure, fine-grained access control for Secrets Manager and S3.

GitHub - Hosts and manages the project code with version control for collaboration.

Architecture Diagram



Clone the original GitHub repository and review the code. Replace hardcoded credentials with a secret in AWS Secrets Manager, update the app to fetch them securely, and push the secure code to your forked repository.

Steps to Implement

1. Clone the Repository:

Fork and clone the original GitHub repository containing the sample web app. This will serve as the base project for adding secure secret management.

2. Review Application Code:

Inspect the cloned code carefully and identify hardcoded credentials (like AWS access keys) that pose major security risks in production.

3. Create AWS Secret:

Log in to the AWS console, open Secrets Manager, and securely store your credentials as a new secret for safe access.

4. Update App Configuration:

Modify the app's config.py (or equivalent file) to dynamically fetch credentials directly from Secrets Manager instead of using hardcoded values.

5. Push Secure Codebase:

Test the updated application thoroughly to confirm it retrieves credentials securely, then commit and push changes to your forked GitHub repository.

Story Time

AWS Secrets Manager (The Secret Keeper's Vault):

AWS Secrets Manager is your digital vault that locks away passwords, API keys, and credentials. Instead of hiding them under your code's pillow, you fetch them securely whenever needed.

Amazon S3 (The Cloud's Treasure Chest):

Amazon S3 is like a bottomless treasure chest where you can store and find your data. Your app just peeks inside to list all the shiny buckets waiting for you.

AWS IAM (The Gatekeeper of Permissions):

IAM is the gatekeeper standing tall, deciding who gets in and what they can touch. It makes sure your app only gets exactly what it's allowed to.

GitHub (The Code Playground):

GitHub is your playground where code lives, evolves, and collaborates. Fork, clone, and push your magical secure code for everyone to learn and share.

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THANK YOU!

